YOUTH CLIMATE RESILIENCE MANDATE

Policy Resolution TG-2026-4309 • District 6 - Central Urban Basin

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Delegation: Lincoln High Youth Environmental Action
Submitted To: City Council
Date: October 24, 2026

-0.9°C (-1.6°F)	4,850 m³	9.80 ha	SDG 15.3.1
HEATWAVE COOLING (INVEST)	DESIGN STORM RUNOFF DIVERTED	GROUND DE-PAVED (5.00 HA ROOFS)	COMPLIANT (AVOIDED ARTIFICIAL DEGRADATION)

FORMAL POLICY CLAUSES

WHEREAS, urban heat islands and severe precipitation events directly impact residents, youth, and neighborhood infrastructure within District 6 - Central Urban Basin;

WHEREAS, empirical biophysical modeling through Stanford University's InVEST urban cooling engine demonstrates that expanding vegetative canopy and cool surfaces can lower local summer surface temperatures by -0.9°C (-1.6°F);

WHEREAS, impervious surface coverage (ESA WorldCover Class 50) exacerbates street-level runoff, and bio-retention interventions modeled by USDA-SCS Curve Number hydrology divert up to 4,850 m³ of runoff during a severe design storm;

WHEREAS, the 5-year simulation scenario completed by Maya Lin (Lincoln High Youth Environmental Action) models verified land cover transitions aligning with UN SDG 15.3.1 Land Degradation Neutrality targets within a $200,000 municipal budget.

NOW, THEREFORE, BE IT RESOLVED, the City Council shall formally adopt a natural infrastructure target to de-pave a minimum of 9.8 hectares of impervious ground and retrofit 5.0 hectares of rooftop area across District 6 - Central Urban Basin;

NOW, THEREFORE, BE IT RESOLVED, municipal capital improvement authorities shall prioritize nature-based solutions, including shade canopy corridors and roadside bioswales, in future roadway and sidewalk projects;

NOW, THEREFORE, BE IT RESOLVED, the City Council establishes a permanent Youth Climate Advisory Seat to review annual infrastructure investments and track local progress toward climate resilience.

ACTIONABLE PUBLIC COMMENT BRIEFING (3 KEY MANDATES)

0:00 - 0:30 1. Neighborhood Heat Mitigation & Surface Cooling

Speaking Note: *"Our empirical simulation demonstrates that with strategic natural infrastructure and surface retrofits, District 6 - Central Urban Basin can achieve a verified -0.9°C (-1.6°F) surface cooling reduction. Grounded in Stanford's InVEST urban cooling model, expanded shade canopy and reflective surfaces reduce solar radiation absorption and increase evaporative cooling along neighborhood corridors."*

Scientific Basis: Stanford Natural Capital Project (InVEST Urban Cooling Model v3.14)

Mandate Demand: Allocate dedicated capital improvement funding to plant shade canopy corridors and cool surfaces across high-heat transit routes in District 6 - Central Urban Basin.

0:30 - 1:00 2. Green Stormwater Infrastructure for Flood Resilience

Speaking Note: *"Our 5-turn planning simulation demonstrates that de-paving obsolete asphalt and installing bio-retention soils diverts 4,850 m³ (1,281,234 gallons) of stormwater during a design storm event. Modeled using USDA-SCS Curve Number hydrology, nature-based infiltration retains water on-site and mitigates street-level runoff."*

Scientific Basis: InVEST Urban Flood Risk Mitigation (USDA Natural Resources Conservation Service NEH-4)

Mandate Demand: Adopt a municipal 'De-Pave First' ordinance prioritizing permeable pavement and roadside bioswales during scheduled roadway and sidewalk capital renewals.

1:00 - 1:30 3. Verifiable Land Degradation Neutrality & Youth Future

Speaking Note: *"By de-paving 9.80 hectares of impervious ground and retrofitting 5.00 hectares of rooftop area, District 6 - Central Urban Basin directly advances Land Degradation Neutrality (LDN) benchmarks under UN SDG Indicator 15.3.1. Our 5-year simulation models verified biophysical trade-offs within municipal budget limits to safeguard neighborhood resilience."*

Scientific Basis: Indicator 15.3.1: Proportion of land that is degraded over total land area & Sub-Indicator 1: Land Cover and Land Cover Change (ESA WorldCover Transition Matrix)

Mandate Demand: Formally incorporate these natural infrastructure targets into the municipal General Plan update and involve youth delegates in capital improvement budget reviews.